Kwan Man (Nick) Cheng

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Vision-Language Learning | Agentic Systems | Robot Learning | Generative Models

EDUCATION

Carnegie Mellon University — **Aug 2026 - May 2028**
M.S. in Robotics — Pittsburgh, PA

University of Wisconsin-Madison — **Jan 2024 - May 2026**
B.S. in Computer Science (transferred from Beijing Normal University), GPA: 3.98/4.00 — Madison, WI
Teaching Assistant, Matrix Methods in Machine Learning | Dean's List (all semesters)

University of California, Berkeley — **Jan 2023 - Jan 2024**
Exchange & Summer Program, GPA: 3.93/4.00 — Berkeley, CA
A+ grades in Data Structures and Introduction to Artificial Intelligence (Top 1%)

RESEARCH & PROFESSIONAL EXPERIENCE

Carnegie Mellon University - Robotics Institute — **Apr 2026 - Present**
Research Assistant | Prof. Fernando De la Torre — Pittsburgh, PA

- Developed SkillWeaver's agentic data-generation and supervision system, integrating VLM-guided Monte Carlo Tree Search, PPO-trained interaction skills, and automated verification.
- Scaled autonomous generation to 11K+ scenes, 31K demonstrations, and 8.7M frames. Augmenting training with generated data increased average LIBERO-PRO-Object OOD success from 47.6% to 78.1%.
- Engineered an automated annotation pipeline that generated 128K spatially grounded reasoning commands from 7K+ fully synthetic trajectories, spanning subgoal and motion-phase supervision for hierarchical VLA policies.
- Trained hierarchical robot policies that integrate VLM reasoning and VLA control using the generated supervision, improving LIBERO-PRO success from below 10% to 70%.

University of Wisconsin-Madison - Video Lab — **Jan 2025 - May 2026**
Research Assistant | Prof. Yin Li — Madison, WI

- Co-first-authored DRIFT, which uses residual flow matching to refine VLM predictions for continuous outputs like robot control and temporal/spatial grounding.
- Trained and evaluated DRIFT with Qwen3-VL-2B and OpenVLA. Qwen3-VL-2B + DRIFT achieved 97.9% average success on LIBERO and 61.5% on SIMPLER WidowX.
- Developed a temporal action localization pipeline for factory videos and delivered a Unity/iOS system for real-time ergonomic risk assessment for General Motors.

Qualcomm — **Sep 2025 - Dec 2025**
Machine Learning Engineer (Capstone Project) — Madison, WI

- Built a multimodal Qwen3-VL presentation coach integrating video, transcripts, face, and pose signals.
- Distilled speech-feedback data and exported Qwen3-VL-2B to ONNX for deployment on Snapdragon NPU.

Tsinghua University — **May 2024 - May 2025**
Research Assistant | Prof. Wenwu Zhu and Prof. Bing Xu — Beijing, China (Hybrid)

- Co-developed Modular-Cam, an LLM-directed video diffusion framework using temporal transformers and modular LoRA camera operators for controllable multi-scene generation.
- Co-developed EVAL, an annual 500 m nighttime-light dataset spanning 1986–2024. Achieved pixel-level R^2 of 0.8088 and reduced county-level RMSE by 48.6% relative to the strongest baseline.

Quwan Technology — **Nov 2024 - Jan 2025**
Algorithm Intern — Guangzhou, China

- Fine-tuned text-to-video models on proprietary datasets and built training data pipelines and parallel inference infrastructure using Python multiprocessing and Docker.

SELECTED PUBLICATIONS

*DRIFT: A Residual Flow Adapter for Decoding Continuous Outputs in Vision-Language Models**. Under review at **ICLR 2027.**
SkillWeaver: Agentic Exploration over Neural Interaction Skills for Scalable Robot Data Generation. **CoRL 2026.**
Modular-Cam: Modular Dynamic Camera-view Video Generation with LLM. **AAAI 2025.**
*An Extended VIIRS-like Artificial Nighttime Light Data Reconstruction (1986-2024)**. **Scientific Data, 2026.**
** Co-first author.*

TECHNICAL SKILLS

Machine Learning: PyTorch, LLMs/VLMs/VLAs, multimodal learning, diffusion models, flow matching, LoRA, reinforcement learning, distributed training, transformers, PPO/DPO/GRPO, DAgger
Robotics & Simulation: Isaac Lab, Isaac Sim, LIBERO, SIMPLER, ROS, Blender, inverse kinematics
Languages & Systems: Python, Java, JavaScript, C/C++, Docker, ONNX, Unity, AWS, GCP